AMPICILLIN

Penicillin allergy

IV in neonates

CHEMICAL NAME (2S,5R,6R)-6-[[(2R)-2-amino-2-phenylacetyl]amino]-3,3-dimethyl-7-oxo-4-thia-1-azabicyclo[3.2.0]heptane-2-carboxylic acid (sodium salt for injection)

CLASS Aminopenicillin (β-lactam antibiotic)

TRADE NAMES Penbritin · Ampicin · Ampicillin sodium injection

THERAPEUTIC USES / INDICATIONS

- Neonatal sepsis and meningitis (with gentamicin); covers *Listeria* and Group B strep.
- *Listeria* meningitis in adults (with others).
- Enterococcal infections (endocarditis with gentamicin).
- Group B streptococcal prophylaxis in labour.

PHARMACODYNAMICS

- Inhibits **cell-wall synthesis** via PBPs → bactericidal.
- Spectrum: streptococci, enterococci, *Listeria*, some *E. coli*, *H. influenzae* (if non-β-lactamase).

ADMINISTRATION

- **IV:** slow injection over 3–5 min or infusion; **IM**.
- Use reconstituted solution promptly (unstable).
- Do not mix with aminoglycosides in same syringe/line (inactivation) — flush between.
- Oral form poorly absorbed (largely replaced by amoxicillin).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea.
- Rash (very common in glandular fever).
- Phlebitis at IV site.

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis.
- *C. difficile* colitis.
- Seizures with very high doses in renal failure.
- Interstitial nephritis (rare).

DOSAGES

- **Adult:** 500 mg–2 g IV every 4–6 h.
- **Meningitis/endocarditis:** 2 g IV every 4 h.
- **Neonates:** 50–100 mg/kg every 8–12 h (age-dependent); meningitis higher.
- **GBS prophylaxis in labour:** 2 g IV, then 1 g every 4 h until delivery.
- Reduce frequency in renal impairment.

CONTRAINDICATIONS & CAUTIONS

- Penicillin hypersensitivity.
- **Caution:** infectious mononucleosis, renal impairment, lymphatic leukaemia.

KEY INTERACTIONS

- **Aminoglycosides:** physically incompatible — give separately.
- **Allopurinol:** ↑ rash.
- **Methotrexate:** ↑ levels.
- **Oral contraceptives:** possible ↓ efficacy (minor).

PHARMACOKINETICS

- **Onset:** IV immediate; peak IM 1 h.
- Crosses inflamed meninges and placenta.
- Excreted unchanged in urine. **t½:** 1 h (longer in neonates).

NURSING MANAGEMENT

- Check allergy; observe after first dose.
- Give aminoglycoside and ampicillin separately; flush line.
- Neonates: dose by weight and age; check calculations twice.
- Monitor IV site, temperature, CBC and cultures.
- Watch for diarrhoea.

PATIENT EDUCATION

- Report rash, itching or breathing difficulty.
- Complete the course.
- Report watery or bloody diarrhoea.

KEY POINTS

- IV aminopenicillin — covers Listeria and enterococci.
- Never mix with aminoglycosides.
- Neonatal sepsis: ampicillin + gentamicin.
- Rash with glandular fever.

SOURCES CHECKED **[110]** Ampicillin 500 mg powder for solution for injection (UK SmPC) **[111]** Ampicillin, oral (US label) **[112]** Ampicillin, IV (US label) *Full references p. 432–457*